25901095 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBIF200 - DATA STRUCTURES & ALGORITHM

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Describe the difference between linear and non-linear data structures. [U] / [CO1]
2. Define index in an array. [K] / [CO3]
3. Relate Multistage Graph? [U] / [CO5]
4. What does the terminator symbol represent in a flowchart? [K] / [CO1]
5. What is meant by a dead node? [K] / [CO5]
6. What is complexity trade off? [U] / [CO2]
7. Describe LIFO advantage in applications. [U] / [CO4]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

8. Write the advantages and limitations of Linear Search. [K] / [CO6]

9. Describe deleting index 1 from array. [U] / [CO3]

10. What is a deterministic algorithm? Explain with an example. [U] / [CO1]

11. Define best-case, worst-case, and average-case performance with suitable examples. [K] / [CO2]

12. Explain deletion when list becomes empty. [U] / [CO3]

13. Describe the Knapsack problem and differentiate between 0/1 Knapsack and Fractional Knapsack. [U] / [CO5]

[5x4 = 20]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

14. Describe the basic operations of queue as an ADT and explain the purpose of each operation. [U] / [CO4]

15. Demonstrate Selection Sort on the list 64, 25, 12, 22, 11 and show each iteration. [U] / [CO6]

16. Describe the working of a linked list with an example and explain the role of the head pointer. [U] / [CO3]

17. Describe the array representation of a queue and discuss the conditions for overflow and underflow. [U] / [CO4]

[10x2 = 20]